

三重积分

$$\sum_{i=1}^n m(\Omega_i) \approx \sum_{i=1}^n \mu(\xi_i, \eta_i, \zeta_i) \Delta V_i$$

$$m(\Omega) = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \mu(\xi_i, \eta_i, \zeta_i) \Delta V_i$$

$$\iiint_{\Omega} f(x, y, z) dV = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta V_i$$

三重积分的计算

体密度为 $f(x, y, z)$ 的空间区域 Ω 的质量

$$\Omega = \{(x, y, z) | z_1(x, y) \leq z \leq z_2(x, y), (x, y) \in D\}$$

$$\iint_D dx dy \int_{\varphi_1(x, y)}^{\varphi_2(x, y)} f(x, y, z) dz$$

$$\int_h^K dz \iint_{D(z)} f(x, y, z) d\sigma$$

例: 计算三重积分 $\iiint_{\Omega} y \, dx \, dy \, dz$, Ω 由 $x=0, y=0, z=0, x+y+z=1$ 生成的立体

先-后=

$$\Omega = \{(x, y, z) \mid 0 \leq z \leq 1-x-y, (x, y) \in D\}$$

$$D = \{(x, y) \mid 0 \leq x \leq 1-y, 0 \leq y \leq 1\}$$

$$\text{原式} = \iint_D dx \, dy \int_0^{1-x-y} y \, dz = \iint_D y(1-x-y) \, dx \, dy = \int_0^1 dy \int_0^{1-y} y(1-x-y) \, dx$$

$$= \frac{1}{2} \int_0^1 y(1-y)^2 \, dy = \frac{1}{2} \int_0^1 (y - 2y^2 + y^3) \, dy = \frac{1}{4} - \frac{1}{3} + \frac{1}{8} = \frac{1}{24}$$

先=后-

对 z 求积分 过 $(0, 0, z)$ ($0 \leq z \leq 1$) 作一垂直于 z 轴的平面...

$$D(z) = \{(x, y, z) \mid 0 \leq x \leq 1-y-z, 0 \leq y \leq 1-z\}$$

$$\text{原式} = \int_0^1 dz \iint_{D(z)} y \, dx \, dy = \int_0^1 dz \int_0^{1-z} dy \int_0^{1-y-z} y \, dx = \int_0^1 dz \int_0^{1-z} \frac{1}{2}(1-z-x)^2 \, dx$$

例: $\iiint_{\Omega} y^2 \, dx \, dy \, dz$ Ω 为锥面 $z = \sqrt{4x^2 + 4y^2}$ 与 $z=2$ 所围立体

$$\Omega = \{(x, y, z) \mid \sqrt{4x^2 + 4y^2} \leq z \leq 2, (x, y) \in D\} \quad D: x^2 + y^2 \leq 1$$

$$\iint_D dx dy \int_{\sqrt{4x^2+4y^2}}^2 y^2 dz = \iint_D y^2 (2 - \sqrt{4x^2+4y^2}) dx dy$$

$$\int_0^{2\pi} d\theta \int_0^1 \rho^2 \sin^2 \theta (2 \rightarrow \rho) \rho d\rho = \frac{\pi}{10}$$

例: 6.3.3 $\iiint_{\Omega} (ax+by+cz) dx dy dz$ Ω 为 $x^2+y^2+z^2 \leq 2z$

Ω 关于 $x=0, y=0$ 对称. $\Rightarrow \iiint_{\Omega} ax dx dy dz = 0$, $\iiint_{\Omega} by dx dy dz = 0$
 $D(z): x^2+y^2 \leq 2z-z^2$

原式 $\iiint_{\Omega} cz \cdot dx dy dz = \int_0^2 cz dz \iint_{D(z)} dx dy$
 $= \int_0^2 cz \cdot 6|D(z)| dz = \int_0^2 \pi cz (2z-z^2) dz = \frac{4}{3} c\pi.$

trick1. 若 $f(x, y, z)$ 缺少 x , 先对 x 积为较方便.

trick2. 若将 x 替换为 $-x$

trick3. $x^2 + y^2 \leq z \leq 1$

$$\iiint_{\Omega} x^2 dx dy dz = \iiint_{\Omega} y^2 dx dy dz.$$

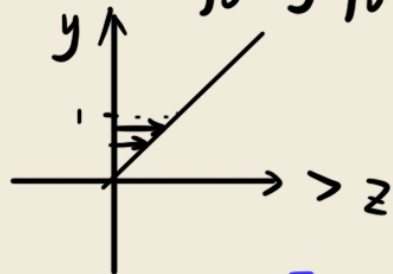
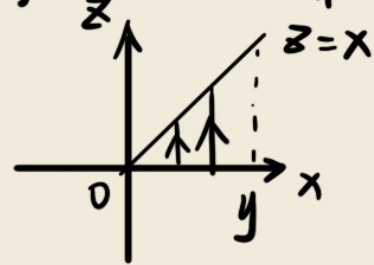
计算累次积分 $\int_0^1 dy \int_0^y dx \int_0^x \frac{e^z}{1-z} dz$

只含有 z , 先对 x, y 求重积分

写出范围 $0 \leq z \leq x \leq y \leq 1 \rightarrow 0 \leq z \leq 1 \quad z \leq x \leq 1 \quad x \leq y \leq 1$

从 z 到 x 到 y . z 不依赖 x, y , x 不依赖 y

例 6.3.4 计算累次积分 $\int_0^1 dy \int_0^y dx \int_0^x \frac{e^z}{1-z} dz$



$$\text{原式} = \int_0^1 dy \int_0^y dz \int_z^y \frac{e^z}{1-z} dx$$

$$= \int_0^1 dy \int_0^y \frac{e^z}{1-z} (y-z) dz$$

$$= \int_0^1 dz \int_z^1 \frac{e^z}{1-z} (y-z) dy$$

$$= \int_0^1 \frac{1}{2} e^z (1-z) dz$$

$$= \frac{1}{2} \int_0^1 (1-z) de^z = \frac{1}{2} (1-z) e^z \Big|_0^1 + \int_0^1 \frac{1}{2} e^z dz = \frac{e}{2} - 1$$

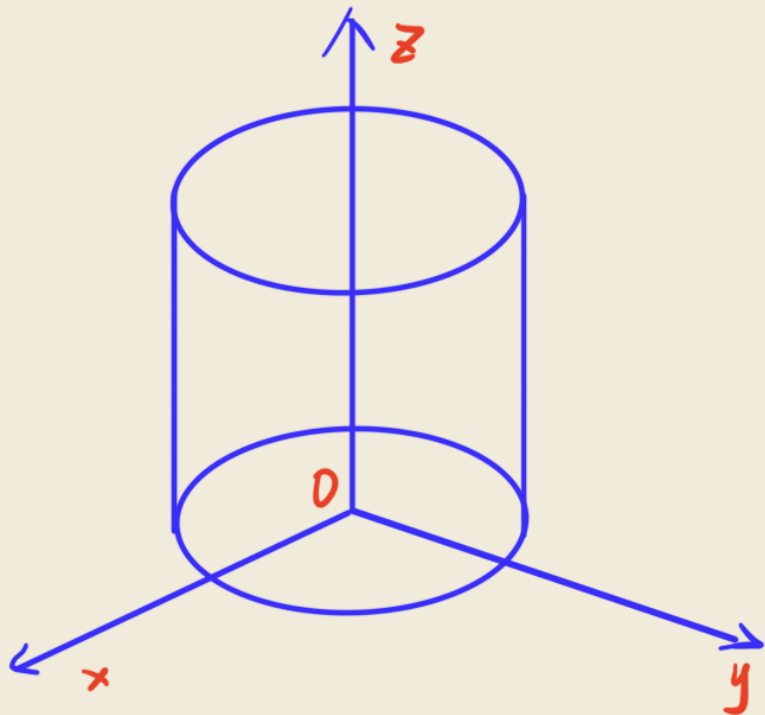
换元积分法

$f(x, y, z)$ 在 Ω 上连续, 取 $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$ 在有界闭区域

Ω' 上连续可微, 使 Ω 与 Ω' 的点一一对应, 雅克比行列式

$$J(u, v, w) = \frac{D(x, y, z)}{D(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} \neq 0$$

柱坐标变换



将 $M(x, y, z)$ 用 (ρ, θ, z) 表示, 其中 (ρ, θ) 是点 M 在 xOy 平面上的投影 P 的

极坐标.

$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \\ z = z \end{cases}$$

$$\text{有 } J(\rho, \theta, z) = \frac{D(x, y, z)}{D(\rho, \theta, z)} = \begin{vmatrix} \cos \theta & -\rho \sin \theta & 0 \\ \sin \theta & \rho \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho$$

旋转性

例 6.3.5 $\iiint_{\Omega} z \, dx \, dy \, dz$, Ω 为 $x^2 + y^2 = z^2$ 与 $z=4$ 所围区域

柱坐标变换 $z = \rho, z=4 \Rightarrow \rho \leq z \leq 4, 0 \leq \rho \leq 4, 0 \leq \theta \leq 2\pi$

$$\text{原式} = \iiint_{\Omega'} \rho z \, d\rho \, d\theta \, dz = \int_0^{2\pi} d\theta \int_0^4 \rho \, d\rho \int_0^4 z \, dz = 64\pi$$

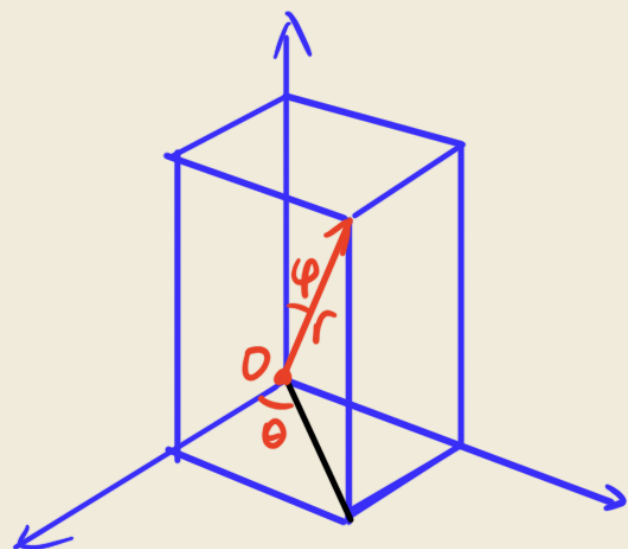
例 6.3.6 $\iiint_{\Omega} z^2 \, dx \, dy \, dz$, Ω 为 $z = x^2 + y^2$ 与 $z = \sqrt{2-x^2-y^2}$ 所围区域

$$\begin{aligned} z &= \rho^2 \quad z = \sqrt{2-\rho^2} \\ \Rightarrow \Omega' &= \{(\rho, \theta, z) \mid \rho^2 \leq z \leq \sqrt{2-\rho^2}, 0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi\} \end{aligned}$$

$$\begin{aligned} \text{原式} &= \iiint_{\Omega'} \rho z^2 \, d\rho \, d\theta \, dz = \int_0^{2\pi} \int_0^1 \rho \, d\rho \int_{\rho^2}^{\sqrt{2-\rho^2}} z^2 \, dz \\ &= 2\pi \int_0^1 \frac{1}{3} \rho \cdot \left[(2-\rho^2)^{\frac{3}{2}} - \rho^6 \right] d\rho = \left(\frac{8\sqrt{2}}{15} - \frac{13}{60} \right) \pi \end{aligned}$$

$$\text{或 } \int_0^{2\pi} d\theta \left(\int_0^1 dz \int_0^{\sqrt{z}} \rho \cdot z^2 d\rho + \int_1^{\sqrt{2}} dz \int_0^{\sqrt{z^2-1}} \rho z^2 d\rho \right)$$

球坐标变换



点 $M(x, y, z)$, 可表示为实数 (r, φ, θ)

$$\begin{cases} x = r \cdot \sin \varphi \cdot \cos \theta \\ y = r \sin \varphi \cdot \sin \theta \\ z = r \cos \varphi \end{cases}$$

$$J(r, \varphi, \theta) = \frac{D(x, y, z)}{D(r, \varphi, \theta)} = \begin{vmatrix} \sin \varphi \cos \theta & r \cos \varphi \cos \theta & -r \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & r \cos \varphi \sin \theta & r \sin \varphi \cos \theta \\ \cos \varphi & -r \sin \varphi & 0 \end{vmatrix} = r^2 \sin \varphi$$

例 0.3.7 $\iiint_{\Omega} z \, dx \, dy \, dz$, Ω 是 $x^2 + y^2 + z^2 \leq 4z$, $\sqrt{x^2 + y^2} \leq z$

$$r^2 \leq 4r \cdot \cos \varphi, \quad r \sin \varphi \leq r \cos \varphi$$

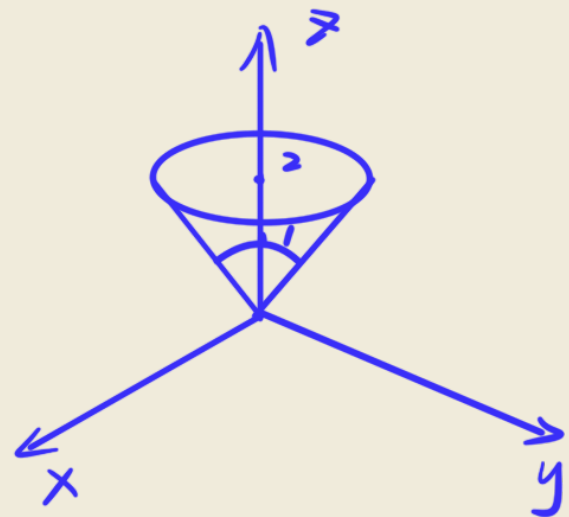
$$\Omega' \left\{ 0 \leq r \leq 4 \cos \varphi, \quad 0 \leq \varphi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi \right\}$$

$$\text{原式} = \iiint_{\Omega'} r^2 \sin \varphi \cdot r \cos \varphi \, dr \, d\varphi \, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{4 \cos \varphi} r^3 \sin \varphi \cos \varphi \, dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \frac{1}{4} (4 \cos \varphi)^4 \cdot \sin \varphi \cos \varphi \, d\varphi = \frac{56}{3} \pi$$

例 6.3.8 Ω 是锥体 $\sqrt{x^2+y^2} \leq z \leq 2$. 计算 $\iiint_{\Omega} |\sqrt{x^2+y^2+z^2}-1| dx dy dz$



$$\Omega_1 \quad x^2+y^2+z^2 \leq 1, \sqrt{x^2+y^2} \leq z \\ 0 \leq r \leq 1, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi$$

$$\Omega_2 \quad x^2+y^2+z^2 \geq 1, \sqrt{x^2+y^2} \leq z \leq 2 \\ 1 \leq r \leq \frac{2}{\cos \varphi}, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi$$

$$\iiint_{\Omega_1} r^2 \sin \varphi (1-r) dr d\varphi d\theta + \iiint_{\Omega_2} r^2 \sin \varphi (1-r) dr d\varphi d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 r^2 \sin \varphi (1-r) dr + \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_1^{\frac{2}{\cos \varphi}} r^2 \sin \varphi (1-r) dr$$

$$= 2\pi \left(\int_0^{\frac{\pi}{4}} \sin \varphi \cdot \frac{1}{12} d\varphi + \int_0^{\frac{\pi}{4}} \sin \varphi \left(\frac{4}{\cos^4 \varphi} - \frac{8}{3} \frac{1}{\cos^3 \varphi} + \frac{1}{12} \right) d\varphi \right)$$

$$= \frac{31\sqrt{2}-30}{6} \pi$$

广义球坐标变换

$$\begin{cases} x = ar \sin \varphi \cos \theta \\ y = br \sin \varphi \sin \theta \\ z = cr \cos \varphi \end{cases}$$

$$J(r, \varphi, \theta) = \begin{vmatrix} a \sin \varphi \cos \theta & a \cos \varphi \cos \theta & -a r \sin \varphi \sin \theta \\ b \sin \varphi \sin \theta & b \cos \varphi \sin \theta & b r \sin \varphi \cos \theta \\ c \cos \varphi & -c r \sin \varphi & 0 \end{vmatrix} = abc r^2 \sin \varphi$$

例 6.3.9 计算三重积分 $\iiint_{\Omega} x^2 dx dy dz$, Ω 是椭球体 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$

$$\begin{cases} x = ar \sin \varphi \cos \theta, & y = br \sin \varphi \sin \theta, & z = cr \cos \varphi \end{cases}$$

$$\Rightarrow r^2 \leq 1, \quad 0 \leq r \leq 1$$

$$\Omega': \quad 0 \leq r \leq 1 \quad 0 \leq \varphi \leq \pi \quad 0 \leq \theta \leq 2\pi$$

$$\text{原式} = \iiint_{\Omega'} a^2 r^2 \sin^2 \varphi \cos^2 \theta \cdot abc r^2 \sin \varphi \, dr d\varphi d\theta$$

$$= a^3 bc \int_0^{2\pi} d\theta \int_0^\pi d\varphi \cdot \sin^3 \varphi \cos^2 \theta \cdot \int_0^1 r^4 \, dr$$

$$= \frac{1}{5} a^3 bc \cdot \int_0^{2\pi} \cos^2 \theta \, d\theta \cdot \int_0^\pi \sin^3 \varphi \, d\varphi \leftarrow \int_0^1 (1-t^2) \, dt$$

$$= \frac{4}{15} a^3 bc \pi$$